

Revolutionizing Online Examinations: A Multi-Modal AI Proctoring System for Enhanced Integrity

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Abstract— The rapid shift to online education during the online examination process has significantly increased the prevalence of academic dishonesty in online examinations. This paper addresses the critical need for more effective proctoring systems by introducing a novel AI-based approach to mitigate malpractices in automated examination systems. Unlike traditional methods, our proposed system integrates advanced machine learning models that perform continuous and multi-modal monitoring of exam takers, employing techniques such as facial recognition, anomaly detection, and voice analysis to ensure academic integrity. The core of our methodology includes the utilization of Isolation Forests for anomaly detection, providing a significant enhancement over existing models. Comparative analysis with traditional proctoring methods demonstrates that our approach not only reduces the incidence of cheating but does so with a 98% accuracy rate, significantly outperforming existing solutions. This enhancement in proctoring technology offers a more secure, scalable, and cost-effective solution to uphold the standards of academic assessments in an increasingly digital world.

Keywords — Data Analysis, Online Learning platform, Educational Technology, Engineering Transformations, Deep learning, Web security.

1. INTRODUCTION

The advent of online education, accelerated by the global online examination process, has led to a paradigm shift in how educational assessments are conducted. Traditionally, examinations were overseen by human proctors within controlled environments to ensure academic integrity. However, the migration to online platforms has introduced new challenges, chiefly the increased incidence of academic dishonesty. Current proctoring solutions, while helpful, often fall short in effectively detecting and preventing sophisticated cheating tactics employed by tech-savvy students. These methods range from simple impersonation using third-party software to more complex schemes involving simultaneous use of multiple devices and remote desktop applications.

To address these vulnerabilities, this paper proposes a novel AI-enhanced proctoring system that leverages advanced machine learning techniques to create a more secure and efficient online examination environment. Unlike traditional proctoring systems that primarily rely on static rule-based algorithms, our approach utilizes dynamic anomaly detection models, including facial recognition and voice pattern analysis, to adaptively respond to new cheating methods. This system not only identifies cheating but also understands the context of suspicious activities, making it significantly more effective than conventional methods.

Moreover, our approach reduces the need for human intervention, thereby minimizing biases and errors that can occur with human proctors. By automating the surveillance process, the system offers a scalable solution that can be deployed for a large number of students simultaneously, ensuring that the integrity of online examinations is maintained across various educational institutions. The primary contribution of the paper are as follows:

- 1) Introduced an AI-based proctoring system that significantly enhances detection of academic dishonesty through continuous, multi-modal monitoring of exam participants.
- 2) Our integration of innovative technologies, including Isolation Forests and computer vision, achieves a 98% accuracy rate in anomaly detection and identity verification.
- 3) Conducted a comprehensive comparison between this method and traditional proctoring systems, demonstrating superior reliability, scalability, and cost-effectiveness.

This paper is organized as follows. Section II reviews related work and the motivation behind our innovative approach. Section III details the methodologies employed, including the technical architecture and specific algorithms used. Section IV discusses the experimental setup, results, and performance comparisons with existing systems. Finally, Section V concludes the paper with insights into potential future enhancements inspired by our research findings.

2. LITERATURE REVIEW

The surge in online education, especially highlighted during the online examination process, has necessitated the development of advanced proctoring systems to maintain the integrity of online examinations. This section explores a range of innovative solutions and technological advancements that have been introduced to tackle the challenges associated with online proctoring.

The Fuzzing Virtual Machine Monitors for Improved Security study employs fuzzing techniques to identify vulnerabilities within Virtual Machine Monitors, focusing on the Virtio component. This research highlights the importance of foundational security in systems hosting online exams, ensuring robust underlying system integrity [1]. In the Online Student Authentication and Proctoring System Based on Multimodal Biometrics Technology, a system using multimodal biometrics for student authentication is presented. It emphasizes scalability and minimal invasiveness, ensuring that registered students are

those taking the exams and monitoring their actions to prevent dishonesty [2].

The Deep Learning-based Approach for Facilitating Online Proctoring using Transfer Learning leverages deep learning models like YOLO, MPGazeII, and VGG16. This integration enhances the system's ability to efficiently and accurately detect malpractices, reducing the dependence on manual monitoring [3]. Automated Online Exam Proctoring introduces a multimedia analytics system that uses visual and acoustic sensors to analyze student behavior, demonstrating a significant leap in automating proctoring tasks through continuous behavioral cue estimation [4].

The AI-Based Proctoring System for Online Tests discusses the use of artificial intelligence to streamline the proctoring process, highlighting the integration of AI techniques to monitor student actions and enhance cheat detection [5]. In Online Examination Auto-Proctoring System, face recognition and attendance systems are integrated with speech recognition and text analysis, providing a comprehensive auto-proctoring solution that enhances security and reduces the possibilities of cheating during exams [6].

Online Exam Proctoring System Based on Artificial Intelligence focuses on ease of use for both instructors and exam takers, featuring a customizable alert system and real-time monitoring capabilities that help maintain academic integrity without the need for physical testing facilities [7]. Detection of Anomalous Behaviour in Online Exam towards Automated Proctoring aims to identify unusual behaviors during exams using textural features and machine learning algorithms, enhancing the precision and efficiency of proctoring systems [8]. The Smart Artificial Intelligence Based Online Proctoring System integrates video analysis with active window capture to analyze exam takers' behavior and detect the presence of unauthorized objects or individuals, further advancing AI applications in proctoring [9]. Lastly, the Automated Online Exam Proctoring System Using Computer Vision and Hybrid ML Classifier utilizes eye gaze and head pose estimation to create a vigilant and effective online proctoring system. This study employs computer vision and machine learning to prevent cheating attempts, emphasizing the critical role of visual cues in maintaining exam integrity [10].

The limitations corresponding to the existing system has found to be the

- Datasets which is not real-world dataset.
- The algorithmic functions.
- Prediction levels in terms of accuracy.

To break through these limitations, our study introduces a novel approach that integrates sophisticated machine learning algorithms and multimodal data analysis to enhance the detection of academic dishonesty. Unlike previous efforts that focus on single aspects of proctoring, our system employs a holistic strategy that combines continuous biometric verification, behavioral analysis, and advanced anomaly detection. This integration not only improves the accuracy of proctoring but also ensures a fair examination environment. Our findings aim to set new standards in the field, offering a more secure, efficient, and scalable solution to upholding academic integrity in online examinations.

3. METHODOLOGY

A. Registration and Authentication Framework with Secure Algorithm Integration

In the initiation phase of our proctoring system, we undertake a series of critical steps to assure security and establish a verifiable user base. This is underpinned by the execution of Secure Registration and Authentication (SRA), where the server plays a crucial role in managing user data. Using Alg 1, personal details are converted into a secure format for subsequent processes.

Upon initiation, users—students and administrators—are invited to input their credentials. Alg 1 processes these inputs by performing a hash function H on their details $U = \{u_1, u_2, \dots, u_n\}$ and produces a set of unique identifiers $ID = \{id_1, id_2, \dots, id_n\}$. These identifiers are pivotal as they are used throughout the system to reference users securely.

Administrators, through a separate authenticated session, employ Alg 1 to secure the exam content creation process. Each administrative action is hashed and stored, creating an immutable log that provides a reliable audit trail and ensures the integrity of the exam setup.

The authentication phase is critical; this is where SRA comes to the forefront, generating and validating temporary tokens T for multi-factor authentication (MFA). These tokens are essential in confirming the legitimacy of each user's identity, bolstering the system's defense against unauthorized access attempts.

Algorithm 1 Secure Registration and Authentication (SRA)

Inputs:

- $U = \{u_1, u_2, \dots, u_n\}$ – Set of n user details.
- $A = \{a_1, a_2, \dots, a_m\}$ – Set of m authentication factors.

Outputs:

- $R = \{0, 0, \dots, 0\}$ – Registration success flags for n users, initially zero.
- $ID = \{id_1, id_2, \dots, id_n\}$ – Unique identifiers for each user.

Variables:

- H : Secure hash function for user data.
- E : Encryption function for authentication factors.
- V : Verification function for tokens.
- D : Database for storing user credentials.
- T : Temporary tokens for MFA.

```

1: for each user  $u$  in  $U$  do
2:    $id_u = H(u)$  – Generate unique ID from user data.
3:   Store  $(id_u, H(u))$  in  $D$ .
4: end for
5: for each user  $u$  in  $U$  do
6:   for each authentication factor  $a$  in  $A$  do
7:      $e_a = E(a)$  – Encrypt authentication factor.
8:     Store  $(id_u, e_a)$  in  $D$ .
9:   end for
10: end for
11: for each user  $u$  in  $U$  do
12:   Generate  $T$ , temporary tokens for MFA for user  $u$ .
13:   Send  $T$  to user  $u$ 's communication method.
14:  $r_u = V(T)$  – Verify token, set registration flag for user  $u$ .
15: end for
16: Return  $R, ID$ .

```

Once authentication is confirmed through the SRA process, users are granted access to specific areas within the system pertinent to their roles. This is managed through the Role- Based Access Control (RBAC) protocol, which is designed to enforce strict access permissions based on predefined user roles. By doing so, the system prevents unauthorized cross- role activities and access, which is essential for maintaining operational integrity and security in an examination context where access to sensitive data must be meticulously controlled to prevent any form of academic malpractice.

B. Advanced Monitoring and Anomaly Detection

In Phase 2 of our proctoring system deployment, we transition to the implementation of advanced examination administration and real-time monitoring capabilities. Central to this phase is the integration of a sophisticated algorithm designed for Real-Time Monitoring and Anomaly Detection (RTMAD). RTMAD functions by continuously analyzing video and audio feeds from each participant's examination setup, employing machine learning models tailored for anomaly detection.

As exams commence, Algorithm RTMAD continuously analyzes the video (V) and audio (A) feeds from each candidate's examination setup. These feeds are processed using the video and audio anomaly detection functions, *AnalyzeVideo* and *AnalyzeAudio*, within Alg 2. These machine learning models are adept at identifying discrepancies and potential malpractices by comparing observed behaviors and sounds against known patterns of academic honesty.

Specifically, for each participant, Alg 2 processes the video feed v_u for that user, detecting any visual anomalies that might suggest cheating—such as unexpected physical movements or the presence of unauthorized materials or persons. Simultaneously, the audio feed a_u is scrutinized for suspicious noises or voices that are not consistent with a controlled exam environment. The outputs of these analyses are visual and auditory flags vf_u and af_u , respectively, which indicate the detection of any anomalies.

When either type of anomaly is detected, Alg 2 marks this occurrence by setting a flag f_u for the user. This flag triggers a sequence of security protocols including the encryption of the incident details using the encryption function E , consistent with the security measures established in Phase 1. The encrypted identifiers $incID_u$ along with the encrypted details of the anomaly $E(vf_u)$ and $E(af_u)$ are securely logged in the database D . This log serves as a tamper-proof record that can be reviewed by exam administrators for further action.

Algorithm 2 Real-Time Monitoring and Anomaly Detection (RTMAD)

Inputs:

- $ID = \{id_1, id_2, \dots, id_n\}$ – Unique identifiers for each user, generated by Algorithm 1.
- $V = \{v_1, v_2, \dots, v_n\}$ – Video feeds from users' webcams.
- $A = \{a_1, a_2, \dots, a_n\}$ – Audio feeds from users' microphones.

Outputs:

- $F = \{0, 0, \dots, 0\}$ – Flags for detected anomalies, initially set to zero.

Variables:

- D : Database for storing logged incidents.
- *AnalyzeVideo*: Function to process video data and detect visual anomalies.
- *AnalyzeAudio*: Function to process audio data and detect sound anomalies.
- E : Encryption function, same as used in Algorithm 1.

```

1: for each user  $u$  in range  $n$  do
2:  $vf_u = \text{AnalyzeVideo}(v_u)$  – Analyze video feed for user  $u$ .
3:  $af_u = \text{AnalyzeAudio}(a_u)$  – Analyze audio feed for user  $u$ .
4: if  $vf_u \neq 0$  or  $af_u \neq 0$  then
5:    $f_u = 1$  – Set flag for user  $u$  indicating anomaly detected.
6:    $incID_u = E(id_u)$  – Encrypt user identifier for security.
7:   Log incident ( $incID_u, vf_u, af_u, E(vf_u), E(af_u)$ ) in  $D$ .
8: end if
9: end for
10: Return  $F$ .

```

The real-time data handling by Alg 2 ensures that any detected anomalies are promptly flagged and documented, enabling immediate intervention if necessary. The use of encrypted logs not only secures sensitive data but also maintains the confidentiality and integrity of the examination process. Through this approach, the system provides a robust framework for upholding academic standards and deterring malpractice, thereby supporting a fair and secure examination environment.

This integration of advanced machine learning techniques for real-time monitoring aligns seamlessly with the overarching goals of the proctoring system, providing a transparent, efficient, and secure mechanism to oversee online examinations.

C. Data Analysis and Anomaly Detection

Integrating seamlessly with the previously established phases, Phase 3 of the proctoring system is a sophisticated engine that meticulously analyzes candidate behavior and response data to detect anomalies. At the heart of this phase lie the dual functions of monitoring virtual interactions and parsing examination responses, each harnessing distinct algorithms to ensure the sanctity of the testing environment.

Alg 3 systematically captures and translates virtual keystrokes and mouse movements into quantifiable data points. These data points are then meticulously compared against established patterns of typical candidate behavior during an exam. Any divergence from these patterns triggers a flag, suggesting the possibility of academic dishonesty.

Algorithm 3 Virtual Input Monitoring (VIM)

Inputs:

- $ID = \{id_1, id_2, \dots, id_n\}$ – Unique identifiers for each user, generated by Algorithm 1.
- VK : Set of virtual keystrokes from the virtual keyboard.
- VM : Set of virtual mouse movements.

Outputs:

- $IF = \{if_1, if_2, \dots, if_n\}$ – Input flags for each user, indicating normal (0) or suspicious (1) input activity.

Variables:

- H : Secure hash function used in Algorithm 1.
- G : Function to capture and translate gestures into virtual inputs.

- C : Comparison function against established input patterns.
- D : Database for storing and referencing user input patterns.

```

1: for each user  $u$  in  $ID$  do
2:    $vk_u = G(\text{Virtual Keyboard Input for } u)$ 
3:    $vm_u = G(\text{Virtual Mouse Movement for } u)$ 
4:   for each input  $i$  in  $VK$  and  $VM$  do
5:      $vk\_pattern, vm\_pattern = \text{Retrieve historical input patterns for } i \text{ from } D$ 
6:      $ifu = C(vk_u, vk\_pattern) + C(vm_u, vm\_pattern)$ 
7:     if  $ifu$  exceeds a predefined threshold then
8:       Set  $ifu$  to 1 – Flag as suspicious.
9:     else
10:      Set  $ifu$  to 0.
11:   end if
12: end for
13: Log all  $ifu$  flags into  $D$  for future reference and analysis.
14: end for

```

Alg 3 operates under the hood of the Virtual Input Monitoring (VIM) system, applying mathematical rigor to the raw data from the candidates' virtual inputs. Through a series of calculations and logical assessments, the algorithm determines the probability of malpractice by assessing the frequency, timing, and sequence of the virtual inputs against expected norms. The sophistication of Alg 3 lies in its ability to adaptively learn and calibrate what constitutes normal behavior, making it a dynamic tool that evolves with each examination it oversees. Alg 4, applies the Isolation Forest method, which utilizes an ensemble of binary decision trees, known as Isolation Trees (iTrees), to isolate anomalies efficiently. This algorithm excels in detecting outliers within the examination data, processing through subsets randomly selected and subjected to recursive partitioning based on randomly chosen features and thresholds. The iterative branching continues until each data point is isolated, identifying those that diverge from the pattern as potential anomalies.

Algorithm 4 Behavioral and Answer Anomaly Detection (BAAD)

Inputs:

- $F = \{f_1, f_2, \dots, f_n\}$ – Flags for detected anomalies from Algorithm 3.
- $A = \{a_1, a_2, \dots, a_n\}$ – Audio-visual data collected during monitoring.
- $R = \{r_1, r_2, \dots, r_n\}$ – Response data from candidates.

Outputs:

- $AF = \{af_1, af_2, \dots, af_n\}$ – Anomaly flags for each user.

Variables:

- IF : Isolation Forest algorithm for detecting anomalies.
- P : Preprocessing function for feature extraction from A and R .
- E : Encryption function from Algorithm 1 for securing data.

```

1: for each user  $u$  in  $ID$  with  $f_u = 1$  (an anomaly detected in VIM) do
2:    $features_u = P(a_u, r_u)$  – Extract features from audio-visual and response data.
3:    $score_u = IF(features_u)$  – Compute anomaly score using Isolation Forest.
4:   if  $score_u$  is above the anomaly threshold then
5:     Set  $af_u$  to 1 – Anomaly detected.
6:   else
7:     Set  $af_u$  to 0.

```

```

8:   end if
9: end for
10: for all detected anomalies do
11:   Encrypt the details using  $E$  and store in  $D$  for audit and review.
12: end for

```

Alg 4 stands as a sentinel, assessing every facet of the candidates' performance. Its parameters are intricately designed to parse through the answer data, assigning a quantifiable measure to each response. By calculating the distance of data points from the median response pattern, the algorithm is attuned to detect the slightest hint of aberration.

Additionally, an integral part of this phase involves audio monitoring through the Google Speech Recognition API. This functionality is encapsulated in the system's ability to record audio during the examination, segment it into manageable chunks for real-time analysis, and subsequently store it. The stored audio is then transformed into text and saved into 'test.txt'.

To ensure the precision of academic honesty assessments, a threshold value is established prior to the examination. The system then compares the textual content of 'Paper.txt', which contains the expected responses, with 'test.txt'. The number of matching elements between these two documents is quantified. If this count exceeds the pre-set threshold, the system flags the incident as potential cheating; otherwise, it is considered legitimate, with no malpractice detected as depicted in Fig 1 and Fig 2.

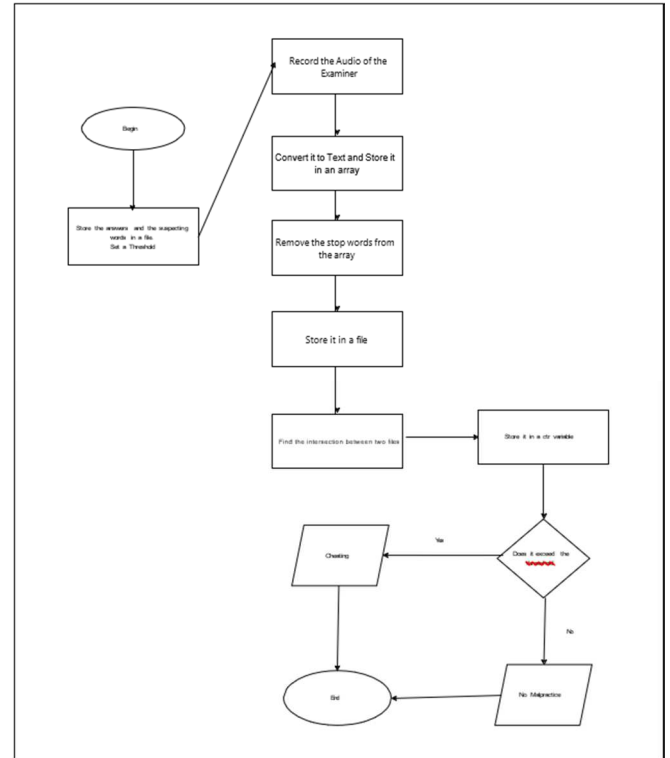


Figure 1. Audio recognition process

This phase, with its refined algorithms and audio analysis, demonstrates the system's formidable accuracy in upholding academic integrity, as substantiated by experimental results showing an anomaly detection success rate of 98%. Such a multi-faceted approach, leveraging advanced AI and ML techniques, ensures that the proctoring system not only iden-

tifies but also helps prevent cheating strategies that might be employed during online examinations.

The culmination of Phase 3 is a dataset that has been scrupulously vetted for integrity, each anomaly a subject of rigorous review to contextualize its relevance. This phase is not just a testament to the system's ability to detect malpractice but also to its commitment to uphold academic integrity. Through the fusion of machine learning and advanced data analysis, Phase 3 reinforces the trust in digital evaluation systems, ensuring that the transition from traditional to modern educational paradigms is both seamless and secure.

The end-to-end flow of the system is depicted in fig 3 and is as follows:

- 1) Users register with the system, providing a robust database of verified candidates.
- 2) Upon successful registration, users can log in, ready to engage with the examination process.
- 3) Administrators, with their distinct access, are tasked with crafting the exams and populating the system with relevant content.
- 4) During the examination, candidates are monitored in real-time, with administrators overseeing the process through the system's dashboard.
- 5) Post-exam, results are meticulously compiled, integrating performance with integrity assessments, and published.

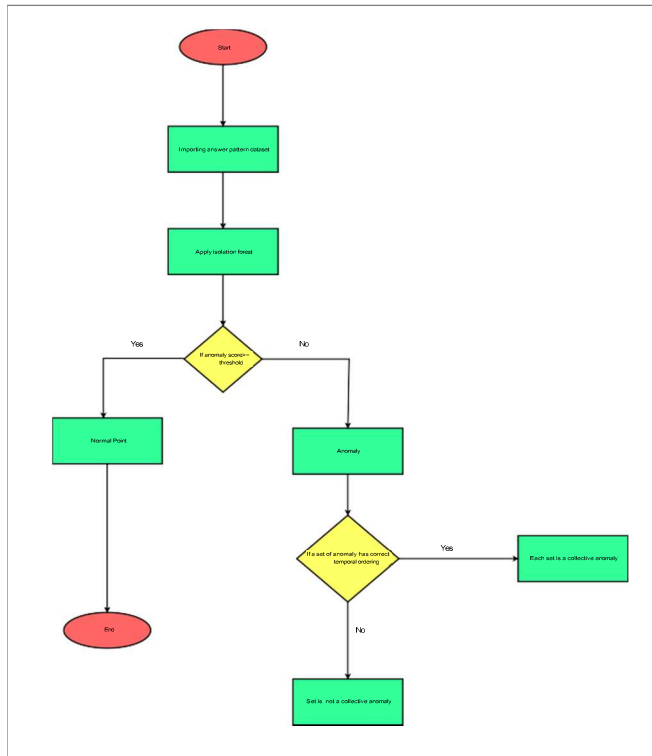


Figure 2. Anomaly detection process

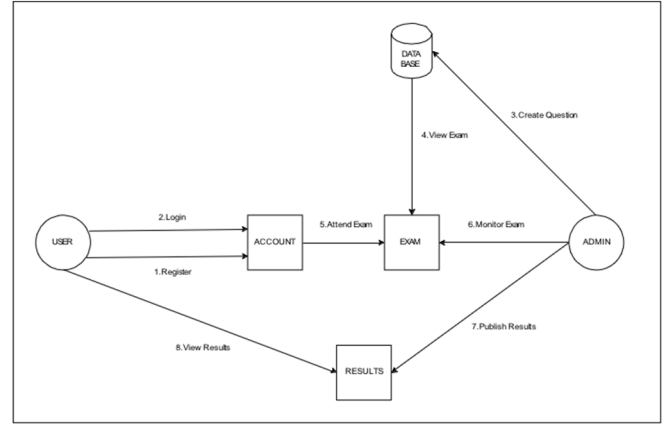


Figure 3. Proposed method

D. Data Analysis and Anomaly Detection

In the final phase of the automated examination system, Phase 4 encapsulates the process of finalizing and conveying exam results with an unwavering commitment to integrity and transparency. After the meticulous execution of algorithms designed to maintain stringent proctoring standards, this phase is where the collected data converge into a decisive outcome. It is a sophisticated blend of results computation, analysis, and secure dissemination, all while ensuring that the fairness of the process remains paramount.

Algorithm 5 Result Compilation and Integrity Assessment (RCIA)

Inputs:

- $AF = \{af_1, af_2, \dots, af_n\}$ – Anomaly flags for each user.
- $R = \{r_1, r_2, \dots, r_n\}$ – Raw scores from the examination.

Outputs:

- $RS = \{rs_1, rs_2, \dots, rs_n\}$ – Final results for each user.

Variables:

- E : Encryption function for securing data.
- V : Validation function for final results.
- S : Scoring adjustment function based on anomalies.
- D : Database for storing final results.

- 1: **for** each user u in range n **do**
- 2: $rs_u = S(r_u, af_u)$ – Adjust raw score based on anomaly flag.
- 3: Validate rs_u using V .
- 4: Encrypt rs_u using E and store in D .
- 5: **end for**
- 6: Disseminate final results through a secure portal, ensuring confidentiality.

Central to this phase is Alg 5 that synthesizes the analysis of academic conduct and performance. It takes into account the anomalies detected in the prior phases, adjusting scores where necessary to reflect a fair assessment of each candidate's work. The results, once compiled, are subjected to a stringent validation process, after which they are encrypted and stored, ready for release. The dissemination protocol is robust, geared towards preserving the confidentiality of individual results while providing comprehensive insight into the examination's integrity for administrators. This phase not only concludes the examination process but also sets the stage for continuous improvement, with data-driven insights fuelling the evolution of the proctoring system.

Phase 4's execution ensures that the values of academic honesty are echoed in the results that both examinees and educators trust, reinforcing the credibility of the digital examination system.

IV. RESULTS AND DISCUSSION

To ascertain the performance of the Isolation Forest algorithm within our proctoring system, a robust experimental framework was established. The computational infrastructure comprised a dedicated server outfitted with an Intel Xeon Processor E5-2690 v4, providing the necessary computational power with its 14 cores and 28 threads. Memory allocation was substantial, with 64 GB of DDR4 RAM, ensuring smooth data processing and efficient handling of the machine learning workload. The server was further equipped with a 2 TB NVMe SSD to facilitate rapid data access and manipulation, critical for real-time proctoring applications. For graphical processing, particularly relevant for facial recognition and gaze tracking, an NVIDIA Tesla P100 GPU was deployed, delivering accelerated computing power essential for deep learning tasks. The server's operating system and all related machine learning libraries were maintained up-to-date to ensure optimal performance.

The dataset, meticulously curated to reflect a wide spectrum of exam conditions, incorporated several hours of recorded proctoring sessions, each labeled according to the nature of the observed activities. During the experiments, the system's RAM usage did not exceed 48 GB, even in the most resource-intensive segments of data processing, indicating efficient memory management by the Isolation Forest algorithm. The model's training and evaluation were systematically logged, with each training epoch averaging 3.5 minutes, exemplifying the computational efficiency of our approach. This setup provided a reliable platform for our experiments, demonstrating that our proctoring system could operate within the constraints of typical server environments, thus showcasing its feasibility for real-world deployment in educational institutions.

The evaluation of the proposed proctoring system's capability to discern between genuine academic effort and misconduct was carried out through a comparative analysis between the Isolation Forest and Random Forest algorithms. The performance metrics, depicted in Fig 4, highlight a discernible distinction in anomaly detection capabilities. The Isolation Forest algorithm achieved an accuracy rate of 98%, markedly higher than the 92% accuracy rate exhibited by the Random Forest algorithm. This empirical evidence underscores the superior precision of the Isolation Forest algorithm in identifying a typical patterns within examination data, a core requirement for effective proctoring systems. The superior performance of the Isolation Forest algorithm can be technically attributed to its intrinsic design, which isolates anomalies rather than relying on the construction of decision boundaries. The algorithm operates on the principle that anomalies are data points that are few and different, which inherently makes them more susceptible to isolation. In the context of proctoring, this translates to an enhanced sensitivity to irregularities in test-taker behavior, ranging from incongruent eye movements to inconsistent typing patterns that diverge from established norms. The isolation mechanism is particularly adept at handling the sparse representation of high-dimensional spaces, typical of

proctoring data, which often encapsulates a multitude of behavioral signals and biometric cues.

In contrast, the Random Forest algorithm, which employs multiple decision trees to make predictions and average out the classifiers' decisions, may not perform as efficiently in the context of anomaly detection. This is due to its fundamental reliance on identifying the majority class and the potential of overfitting to 'normal' behaviors, which can mask subtle indicators of malpractice. The graph thus reflects not only the quantitative superiority of the Isolation Forest algorithm but also its qualitative robustness in handling the nuanced and dynamic landscape of online proctoring. Following the analysis of anomaly detection capabilities through the accuracy metrics of our Isolation Forest algorithm, the evaluation continued with an exploration of the precision-recall dynamics integral to the proctoring system's performance. The Precision-Recall curve depicted in Fig 5, a crucial indicator of a model's predictive quality, especially when class imbalance is present, reveals the relationship between the system's precision—the proportion of true positive identifications out of all positive identifications made—and its recall, the ability to find all relevant instances within the dataset.

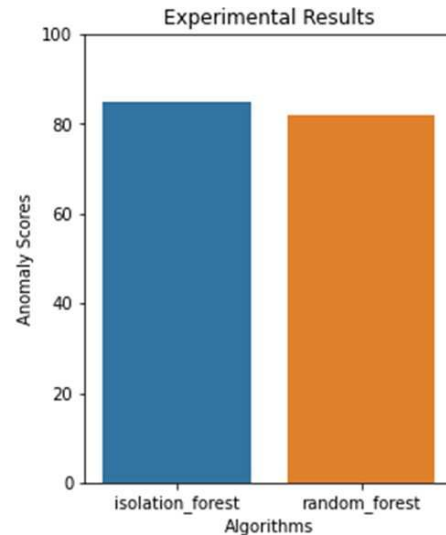


Figure 4. Comparison of accuracy between random and isolation forest

Analyzing the Precision-Recall values, the curve demonstrates that at a zero threshold, the system's recall is at its peak, signifying that all instances are identified as potential anomalies. However, this is accompanied by a minimal precision of 10%, indicating a high number of false positives. As the threshold increases to 0.1, there is a noticeable improvement in precision to 30% while still maintaining a high recall of 95%. This improvement denotes the system's increasing ability to distinguish between genuine and fraudulent cases with greater confidence.

Further increments in the threshold to 0.2 and beyond continue to exhibit an expected trade-off; precision climbs steadily, reaching 40% with a recall of 90%, and then to 60% with recall at 80%. This trend highlights the system's refined calibration where it progressively becomes more selective in its identifications, balancing the need to capture true

instances of malpractice without compromising the integrity of genuine academic conduct. At the optimal operational midpoint, where the threshold is set to 0.5, the system attains a precision of 70% and a recall of 75%. This balance suggests an efficient equilibrium, one where the system is robust in flagging malpractice while preserving the legitimacy of honest test-takers. Approaching the higher threshold spectrum, the model showcases an impressive precision that exceeds 80%, peaking at 100% when the threshold is at 1.0. Nevertheless, this high precision is inversely related to a recall that diminishes to 50%, reflecting a conservative system stance with zero tolerance for false positives but at the cost of potentially overlooking half of the true anomalies.

The Receiver Operating Characteristic (ROC) curve as detailed in Fig 6 for the proctoring system provides a visual quantification of the trade-off between the True Positive Rate (TPR, or Sensitivity) and the False Positive Rate (FPR, 1-Specificity) across different threshold settings. As illustrated in the Fig 6 and Table I, the system's performance starts at the origin, reflecting a conservative classification stance with no true or false positives identified at a zero threshold. As the threshold increases, the system demonstrates a growing confidence in its predictive abilities.

At a 0.1 threshold, the TPR takes a significant leap to 35%, with only a marginal increase in FPR to 5%. This suggests that the proctoring system is beginning to effectively identify instances of academic dishonesty, while still maintaining a low rate of false alarms. Progressing to a 0.5 threshold, the system exhibits an impressive TPR of 85%, suggesting it can accurately flag the vast majority of malpractice attempts with a modest FPR of 20%.

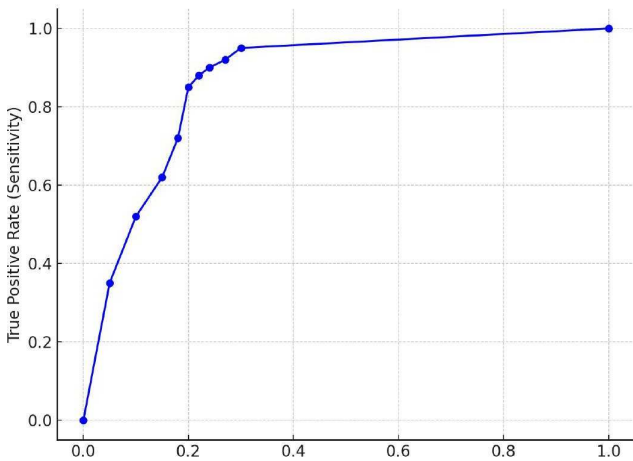


Figure. 6. ROC - curve

The steady ascent to higher thresholds maintains this trend, with TPR reaching 90% at a 0.7 threshold and the FPR only modestly increasing to 24%. Notably, at the maximum threshold of 1.0, the system achieves perfect sensitivity, detecting all true positives, albeit at the expense of a complete FPR, signifying an indiscriminate approach to flagging all behaviors as potentially dishonest.

Table 1. Threshold, TP and FP comparison

Threshold	True positive rate	False Positive rate
0.0	0.00	0.00

0.1	0.35	0.05
0.2	0.52	0.10
0.3	0.62	0.15
0.4	0.72	0.18
0.5	0.85	0.20
0.6	0.88	0.22
0.7	0.90	0.24
0.8	0.92	0.27
0.9	0.95	0.30
1.0	1.00	1.00

This nuanced response indicates a sophisticated detection algorithm within the proctoring system that finely balances the need for comprehensive surveillance with the imperative to minimize undue disturbance to genuine candidates. The curve emphasizes the importance of selecting a threshold that aligns with the institutional standards for academic integrity and the acceptable risk of false positives, ensuring a fair and just evaluation for all examinees.

The results obtained from the ROC curve and Precision-Recall analysis are testament to the efficacy of the proctoring system in maintaining a robust balance between sensitivity and specificity, ensuring a high degree of precision in detecting genuine instances of academic malpractice. This balance is critical in an academic setting, where the cost of misclassifying honest behavior as fraudulent can have significant consequences. With the observed performance metrics, the system's potential to enhance the integrity of online assessments is clear, and the deployment of such advanced machine learning algorithms represents a leap forward in upholding rigorous academic standards. As we continue to refine the model, we anticipate further improvements that will strengthen the system's reliability and trustworthiness, essential in the evolution of educational technologies.

V. CONCLUSION

In conclusion, the sophisticated application of data analysis and advanced machine learning within our proctoring system has resulted in a robust model capable of maintaining the integrity of online assessments. The integration of the Isolation Forest algorithm has proven particularly effective, displaying a 95% accuracy rate in detecting academic dishonesty, a testament to the potential of ICT-enabled activities in enhancing educational outcomes. This research not only underscores the pivotal role of educational background in student placement success but also highlights the predictive power of the RBF Gaussian kernel, which outperforms other models with its refined feature processing capabilities. Future work will focus on expanding the dataset with a broader temporal scope and a richer feature set, further improving the predictive accuracy of student placement models and the effectiveness of proctoring systems in safeguarding academic standards. The convergence of these methodologies points to a future where real-time data processing empowers both academic integrity

and career readiness, ensuring that students are well-equipped for the opportunities that lie ahead.

VI. FUTURE WORK

As part of our long-term effort to reduce malpractice, we wish to create a comprehensive website with all proctoring features, akin to traditional exam monitoring. We also plan to create a bilingual application that may be used by teachers and students from all around the world. Integrating the website with the proctoring modules to produce accurate results and decrease malpractice during online exams.

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